MILITARY SPECIFICATION

# CARTRIDGE, 40MM, M918 TP LOADING, ASSEMBLING AND PACKAGING

This Specification is approved for use within the U.S. Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packaging for one type of cartridge designated as Cartridge, 40mm, M918 TP.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified (see 6.2), the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement there to, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation - Packaging, Methods of
MIL-A-48078 - Ammuniton, Standard Quality Assurance Provisions, General Specification for
MIL-P-60942 - Primer Percussion for 40mm Ammunition

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

**Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.**


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2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

DRAWINGS (see 6.11)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

PRODUCT AND PACKING DRAWINGS

| | |
|---|---|
| 8886405 | - Linking of 40mm Hi Velocity Ammunition |
| 9251995 | - Box, Wirebound, Packing Ammunition for Cartridge, 40mmm Linked |
| 9251996 | - Box, Fiber, Packing, Ammunition for Cartridge, 40mm Linked |
| 9399372 | - Cartridge, 40mm, M918TP |
| 9362543 | - Packing and Marking of Shipping and Storage Container M548 with Linked 40mm Cartridges |

INSPECTION EQUIPMENT DRAWINGS

| | |
|---|---|
| 9202253 | - Chamber Gage |
| 9202254 | - Limit |
| 9202255 | - Alignment |
| 9202929 | - Action Time |

CODE OF FEDERAL REGULATIONS

| | |
|---|---|
| TITLE 49 | - Transportation, Parts 100-199 |

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify "49 CFR 100-199 (latest revision)).

(Copies of specifications, standards, handbooks and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)



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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless specified exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Cartridge. The cartridge shall comply with all requirements specified on Drawing (Dwg.) 9399372, all associated drawings and with all requirements in applicable specifications.

3.3 X-ray examination of projectile assembly prior to assembly to cartridge case assembly. The projectile assembly shall be x-rayed for improper assembly, missing or damaged parts, missing operations, foreign material and an armed or partially armed fuze.

3.4 X-ray examination of ballistic sample. Prior to forwarding the cartridges to the proving ground for ballistic testing, they shall be subjected to x-ray examination.

3.5 Functioning. The cartridge assembly shall function without casualty and comply with the following:

3.5.1 Velocity. The average velocity of the sample cartridges, conditioned at 70°F and fired in the MK19, MOD III, Grenade machine gun shall be 790 ± 10 feet per second (fps) with a standard deviation of 12.0 fps or less.

3.5.2 Action time. When fired in the M129 Grenade Launcher, the action time of the cartridge, conditioned at 70°F, shall not exceed 4 milliseconds.

3.5.3 Impact functioning. When fired from a MK19 MOD III, Grenade Machine Gun or M129 Grenade Launcher, the projectile assembly upon impact shall function satisfactorily as evidenced by smoke, flash and sound. There shall be no evidence of an air burst (see 6.9).

3.5.4 Safety. When fired from the MK19, MOD III, Grenade Machine Gun or M129 Grenade Launcher, the projectile shall not lodge in the bore of the weapon and there shall be no evidence of a premature burst (see 6.10).



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3.6 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough and workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however a few light scraches not exposing base material may be permitted. All marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Test equipment and inspection facilities. The manufacturer shall insure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality Conformance Inspection (4.4)

4.3 First article inspection. Not applicable.

4.4 Quality conformance inspection.



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4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Each inspection lots shall contain:

a. Cartridge case assembly metal parts from one lot interfix number and from one manufacturer.

b. Primers from one lot interfix number and from one manufacturer.

c. Projectile assemblies from no more than two (2) consecutively produced serial lots and from one manufacturer.

d. Propellant from not more than one (1) lot number and from one manufacturer.

e. Cartridges in less than full belt quantities shall be handled in the following manner:

(1) If the remaining quantity exceeds 25 cartridges, use cartridges from next lot to complete the belt.

(2) If the remaining quantity is less than 25 cartridges include cartridges in next belt produced.

(3) Packing box and data cards shall be annotated as to the mixture.

f. Links from one lot interfix number and one manufacturer.

4.4.2 Examination for defects.

a. Sampling plan. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.



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## CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48684(AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 8886327 |
|---|---|---|---|---|---|
| 4.4.2.1 | Cartridge Case Assembly, Prior to Loading | | | | NEXT HIGHER ASSEMBLY 9399372 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| CRITICAL | | | | | |
| 1. | Closing cup missing | | 100% | 3.2 | Visual/Probe |
| MAJOR | | | | | |
| 101. | Propellant weight | | | 3.2 | 4.5.2 |
| 102. | Propellant weight (alternate method) | | | 3.2 | 4.5.3 |
| MINOR | None defined | | | | |

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.



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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48684(AR)

| PARAGRAPH | TITLE | | SHEET OF | DRAWING NUMBER |
|---|---|---|---|---|
| 4.4.2.2 | Cartridge Case Assembly, Prior to Insertion of Base Plug and Primer | | 1 1 | 8886327 |
| | | | | NEXT HIGHER ASSEMBLY 9399372 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH / PARAGRAPH REFERENCE /INSPECTION METHOD |
| CRITICAL | | | | |
| 1. | Spilled propellant from cartridge case | | 100% | 3.2 Visual |
| 2. | Propellant charge weight obviously incorrect (See 6.8) | | 100% | 3.2 Visual |
| MAJOR | | | | |
| 101. | Propellant charge weight obviously incorrect (See 6.8) | | 100% | 3.2 Visual |
| MINOR | None defined | | | |

NOTES:

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## CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 8886327 |
|---|---|---|---|---|---|
| 4.4.2.3 | Cartridge Case Assembly | | | | NEXT HIGHER ASSEMBLY 9399372 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| CRITICAL | | | | | |
| 1. | Primer above flush with base plug | | 100% | 3.2 | Gage |
| MAJOR | | | | | |
| 101. | Security of crimp of base plug and action time | 1/ | 2/ | 3.2<br>3.5.2 | 4.5.4<br>9202929 |
| 102. | Base plug above flush | | .40 | 3.2 | Gage |
| 103. | Depth of base plug, max | | .40 | 3.2 | Gage |
| 104. | Depth of primer from base plug, max. | | .40 | 3.2 | Gage |
| 105. | Primer damaged | | .40 | 3.2 | Visual |
| 106. | Excessive sealing compound on exterior surface of primer | | .40 | 3.2 | Visual |
| MINOR | | | | | |
| 201. | Poor Workmanship | | .65 | 3.7 | Visual |

NOTES: 1/ Five (5) cartridge case assemblies shall be randomly selected every four (4) hours from each manual primer assembly machine or twenty (20) cartridge case assemblies shall be randomly selected every four (4) hours from each automatic primer assembly machine.
2/ If any base plug moves more than the applicable requirment or if any action time is greater than the applicable requirement, the four (4) hour production quantity represented shall be rejected.





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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48684(AR)

| PARAGRAPH | TITLE | SHEET | OF | DRAWING NUMBER |
|---|---|---|---|---|
| 4.4.2.4 | Projectile Assembly Cartridge, 40mm M918TP | 1 | 1 | 9399374 |
| | | | | NEXT HIGHER ASSEMBLY 9399372 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| CRITICAL 1. | X-ray examination 1/ | | 100% | 3.3 | 4.5.5 |
| MAJOR 101. | X-ray examination 2/ | | 100% | 3.3 | 4.5.5 |
| MINOR 201. | X-ray examination 2/ | | 100% | 3.3 | 4.5.5 |

NOTES: 1/ A critical defect is defined as a setback pin not adjacent to the rotor gear, setback pin not seated in bottom of counterbore in escapement housing, setback pin or spring missing, detent spring or detent missing, detent not engaged in teeth of rotor gear, or pinion or verge assembly missing.
2/ A major or minor defect is defined as any defect not specified in Note 1/.



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## CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48684(AR)

| PARAGRAPH | TITLE | SHEET 1 OF 1 | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.5 | Projectile Assembly, priorto assembly to Cartridge Case Assembly | | 9399374 |
| | | | NEXT HIGHER ASSEMBLY 9399372 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| CRITICAL | None defined | | | | |
| MAJOR 101. | O-ring (packing) missing from groove | | .40 | 3.2 | Visual |
| MINOR | None defined | | | | |

NOTES:

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Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.



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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48684(AR)

| PARAGRAPH | TITLE | SHEET 1 OF 2 | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.6 | Cartridge, 40mm, M918TP | | 9399372 |
| | | | NEXT HIGHER ASSEMBLY |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| CRITICAL | | | | | |
| 1. 1/ | X-ray examination of ballistic sample | 432 | ACC-0<br>REJ-1 | 3.4 | 4.5.6 |
| MAJOR | | | | | |
| 101. | Chamber gage failure | | 100% | 3.2 | 9202253<br>9202254<br>9202255 |
| 102. | Total length, maximum | | 100% | 3.2 | Gage |
| 103. 2/ | X-ray examination of ballistic sample | 432 | 3/ | 3.4 | 4.5.6 |
| 104. | Air pressure test | 200 | ACC-7<br>REJ-8 | 3.2 | 4.5.7 |
| 105. | Pull test of projectile | 80 | ACC-1<br>REJ-2 | 3.2 | 4.5.8 |
| 106. | Gap, maximum, at rear of rotating band | | .40 | 3.2 | Gage |
| 107. | Packing exposed | | .40 | 3.2 | Visual |

NOTES: 1/ A critical defect is defined as any defect listed in Note 1/ of Paragraph 4.4.2.4.
2/ A major or minor defect is defined as any defect not specified in Note 1/.
3/ Major or minor defects found during inspection shall be counted against the sample that was selected for detection of that particular defect. Any sample containing a known defect shall not be fired.



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## CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | SHEET 2 OF 2 | DRAWING NUMBER 9399372 |
|---|---|---|---|
| 4.4.2.6 | Cartridge, 40mm, M918TP | | NEXT HIGHER ASSEMBLY |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| MINOR | | | | | |
| 201. 2/ | X-ray examination of ballistic sample | 432 | 3/ | 3.4 | 4.5.6 |
| 202. | Marking missing, improper or illegible | | .65 | 3.2 | Visual |
| 203. | Poor Workmanship | | .65 | 3.7 | Visual |
| | Functioning Tests | | | | |
| | Functioning | | | | |
| 4/ | MK19, MOD III, machine gun - single shot | 144 | 4/ | 3.5, 3.5.1, 3.5.3, 3.6 | 4.5.9.1 |
| 4/ | MK19, MOD III, machine gun, rapid fire | 144 5/ | 4/ | 3.5, 3.5.3, 3.5.6 | 4.5.9.2 |
| 4/ | M129 Grenade launcher | 144 | 4/ | 3.2, 3.5.2, 3.6 | 4.5.9.3 |

NOTES: 4/ See Table I for defect classification and acceptance/rejection criteria.
5/ Three (3) full metal cans (48 linked cartridges per can) shall be randomly selected.



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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48684(AR)

| PARAGRAPH | TITLE | SHEET 1 OF 1 | DRAWING NUMBER 8886405 |
|---|---|---|---|
| 4.4.2.7 | Linked Ammunition, Prior to Packing | | NEXT HIGHER ASSEMBLY |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| CRITICAL 1. | Linked improperly (anyone of the four (4) tabs of link not in groove; link inverted; link damaged; link distorted) | | 100% | 3.2 | Visual |
| MAJOR | None defined | | | | |
| MINOR | None defined | | | | |

NOTES:

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## CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48684(AR)

| PARAGRAPH | TITLE | | SHEET OF | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.8 | Unsealed Fiber Box or Unsealed Shipping and Storage Container | | 1 1 | | 9251996 or 9362543 |
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| CRITICAL | None defined | | | | |
| MAJOR | | | | | |
| 101. | Cartridge inverted | | .40 | 3.2 | Visual |
| 102. | Incorrect number of cartridges | | .40 | 3.2 | Visual |
| 103. | Fillers missing | | .40 | 3.2 | Visual |
| 104. | Support missing | | .40 | 3.2 | Visual |
| 105. | Spacer missing (when applicable) | | .40 | 3.2 | Visual |
| 106. | Top pad missing (when applicable) | | .40 | 3.2 | Visual |
| MINOR | None defined | | | | |

NOTES:

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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48684(AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 9251996 |
|---|---|---|---|---|---|
| 4.4.2.9 | Sealed Fiber Box, Prior to Sealing Bag | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| CRITICAL | None defined | | | | |
| MAJOR | | | | | |
| 101. | Container damaged | | .40 | 3.2 | Visual |
| 102. | Corners not blunted | | .40 | 3.2 | Visual |
| 103. | Heat seal test of packing seals | 1/ | 1/ | 3.2 | 4.5.10 |
| MINOR | | | | | |
| 201. | Contents loose | | .65 | 3.2 | Manual |
| 202. | Label missing or unidentifiable | | .65 | 3.2 | Visual |

NOTES: 1/ This test shall be performed after sealing bag. Sampling and rejection shall be in accordance with MIL-P-116.

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## CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 9251995 |
|---|---|---|---|---|---|
| 4.4.2.10 | Unsealed Wooden Packing Box | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| CRITICAL | None defined | | | | |
| MAJOR | | | | | |
| 101. | Marking of bag misleading or unidentifiable | | .40 | 3.2 | Visual |
| 102. | Bag torn or perforated | | .40 | 3.2 | Visual |
| 103. | Bag improperly sealed | | .40 | 3.2 | Visual |
| MINOR | None defined | | | | |

NOTES:





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## CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48684(AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 9251995 or 9362543 |
|---|---|---|---|---|---|
| 4.4.2.11 | Sealed Wooden Packing Box or Sealed Shipping and Storage Container | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| CRITICAL | None defined | | | | |
| MAJOR | | | | | |
| 101. | Box or container damaged | | .40 | 3.2 | Visual |
| MINOR | | | | | |
| 201. | Contents loose | | .65 | 3.2 | Manual |
| 202. | Car seal missing or improperly positioned | | .65 | 3.2 | Visual |
| 203. | Marking misleading or unidentifiable | | .65 | 3.2 | Visual |

NOTES:

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TABLE I - QUALITY CONFORMANCE INSPECTION

FUNCTIONING & DEFECT CLASSIFICATION

AND ACCEPTANCE/REJECTION CRITIERIA

MK19, MOD III, GRENADE MACHINE GUN & M129 GRENADE LAUNCHER

| Defect | Acceptance Number | | | Rejection Number | | | Classification |
|---|---|---|---|---|---|---|---|
| | MK19 Single Shot | MK19 Rapid Fire | M129 | MK19 Single Shot | MK19 Rapid Fire | M129 | |
| 1. Failure to meet Average velocity (see 3.5.1) | 1/ | -- | -- | 2/ | -- | -- | Major |
| 2. Standard deviation of velocity over max. (see 3.5.1) | 1/ | -- | -- | 2/ | -- | -- | Major |
| 3. Action time over max (see 3.5.2) | -- | -- | 1 | -- | -- | 2 | Major |
| 4. Impact functioning failure (see 3.5.3) | 10 | 10 | 10 | 11 | 11 | 11 | Major |
| 5. Primer leak as evidenced by perforation of primer cup or smoke deposit on base plug or cartridge case (see 3.5) | 0 | 0 | 0 | 1 | 1 | 1 | Major |
| 6. Projectile lodges in weapon bore (see 3.5.4) | 0 | 0 | 0 | 1 | 1 | 1 | Critical |
| 7. Premature burst (see 3.5.4) | 0 | 0 | 0 | 1 | 1 | 1 | Critical |



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TABLE I - QUALITY CONFORMANCE INSPECTION

FUNCTIONING & DEFECT CLASSIFICATION

AND ACCEPTANCE/REJECTION CRITIERIA

MK19, MOD III, GRENADE MACHINE GUN & M129 GRENADE LAUNCHER

| Defect | Acceptance Number | | | Rejection Number | | | Classification |
|---|---|---|---|---|---|---|---|
| | MK19 Single Shot | MK19 Rapid Fire | M129 | MK19 Single Shot | MK19 Rapid Fire | M129 | |
| 8. Airburst (see 3.5.3) | 0 | 0 | 0 | 1 | 1 | 1 | Major |
| 9. Base plug more than .005 above flush with rear of cartridge case after firing (see 3.2) | -- | -- | 2 | -- | -- | 3 | Major |
| 10. Base plug more than .010 above flush with rear of cartridge case after firing | -- | -- | 0 | -- | -- | 1 | Major |
| 11. Primer misfire (see 3.5) | 1 | 1 | 1 | 2 | 2 | 2 | Major |
| 12. Weapon stoppage excluding primer misfire (see 3.5) | 0 | 0 | 0 | 1 | 1 | 1 | Major |

NOTES:

1/ Lot acceptable if requirement is met

2/ Lot rejected if requirement is not met.



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4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Check test for deterioration of primers. If the total time between original acceptance of any lot and the assembly of that lot into the cartridge exceeds two years, or if the primers have been subjected to adverse conditions; however brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-P-60942 immediately before the primer lot is assembled into the cartridge. This test shall be performed by the contractor on primers selected by the Government inspector at the facility assembling the primers into the cartridge (see 6.6).

4.5.2 Propellant weight in cartridge case. The propellant weight shall be determined and then check weighed 100 percent. The check weighing shall be accomplished independently of the original weighing for determination, using a different balance from that used to make the original weighing and if performed manually, shall be performed by another operator. Any charge which fails to comply with the requirement specified on the applicable drawing shall be classified defective and removed from the lot.

4.5.3 Propellant weight in cartridge case, alternate method. At the start of production, 380 consecutive samples per station per machine shall be selected and weighed 100 percent. All samples shall comply with the applicable drawing requirement. If all samples comply with the applicable drawing requirement, five (5) samples per station from every 2 hours production shall be selected and the average weight of the five (5) samples must meet the assessed mean propellant weight within $\pm$ 30milligrams. If the average weight fails the applicable drawing requirement, correction will be made and 380 consecutive samples weighed from that station. If all samples meet the applicable requirement, the sampling every 2 hours will be resumed.

4.5.4 Security of crimp of base plug and action time. The cartridge assembly shall be fired in a M129 Grenade launcher. Observation shall be made for base plug movement and action time. The movement of the base plug shall be measured from the highest point of the base plug to a point on the cartridge case adjacent to it.



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4.5.5. X-ray examination of projectile assemblies. Prior to assembly to the cartridge case assembly, the projectile assembly shall be x-rayed for critical, major and minor defects. The test shall be conducted in two (2) planes. One shall be from the top of the projectile down and the other from the side.

4.5.6 X-ray examination of ballistic sample. Prior to the fowarding of the cartridges to the proving ground for ballistic testing, the cartridges shall be x-rayed for critical, major and minor defects. The test shall be conducted in two (2) planes. One shall be from the top of the cartridge down and the other from the side.

4.5.7 Air pressure. The cartridge assembly shall be placed in an approved fixture and a measured quantity of air shall be applied to produce the required air pressure.

4.5.8 Pull test of projectile. The cartridge shall be placed in an approved fixture and the axial tension specified on the applicable drawing shall be applied. Cartridge case assemblies may be reused. The projectile shall be pulled until separation occurs and data recorded.

WARNING: This test is conducted on assemblies containing an explosive element. Test barricades (where used), procedures and equipment shall have prior approval.

4.5.9 Functioning. This test shall be conducted at a Government owned proving ground. The cartridge being tested shall be temperature conditioned at 70° ± 5°F for a minimum of eight (8) hours, and then fired within fifteen (15) minutes after being removed from temperature soak. The weapon used shall have been "broken in" if new with at at least forty (40) cartridges. At the start of the functioning test, six (6) gun warmers shall be fired.

4.5.9.1 MK19, MOD III, grenade machine gun, single shot. The linked cartridges shall be fired single shot and impact at 200 meters ± 5 meters from the weapon into 10 inches, minimum, of coarse sand. The cartridges shall be fired at a rate not to exceed four per minute. The sand shall be raked so that individual craters do not overlap. In addition, the sand shall be raked after each dud. If a projectile assembly falls outside the impact medium, it shall be declared a "no-test". The velocity shall be measured at 30 feet from the muzzle of the machine gun. The monitoring devices and the recorders shall be a dual system setup for recording two (2) separate readings and shall be within 3 fps of each other. If they are not within 3 fps the results shall be discarded, the cartridge declared "no-test" and another fired in its place. If a weapon is suspected of being the cause of rejection for the mean velocity and standard deviation phase, sufficient cartridges from a controlled lot shall be fired from the same approved weapon. If there is a significant difference in



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the standard deviation and mean velocity of the control lot, then the lot shall be rejected. Observation shall be made for functioning at the impact medium and any premature burst or air burst. Velocity shall be recorded. The standard deviation of velocity shall be calculated. The expended cartridge case shall be examined for primer leakage. When the proving ground reports ballistic test results, the number of "no-test" cartridges shall also be recorded.

4.5.9.2 MK19, MOD III grenade machine gun, rapid fire. The linked cartridges shall be fired in bursts of five (5), and impact at 200 meters ± 5 meters from the weapon into 10 inches, minimum, of coarse sand. The sand shall be raked so that individual craters do not overlap. In addition, the sand shall be raked after each dud. If a projectile assembly falls outside the impact medium , it shall be declared a "no-test". Velocities of the cartridge shall not be recorded. Observation shall be made for functioning at the impact medium and any premature bursts or airburst. The expended cartridge cases shall be examined for primer leakage. When the proving ground reports ballistic test results, the number of "no-test" cartridges shall also be recorded.

4.5.9.3 M129 grenade launcher. The linked cartridges shall be fired from the grenade launcher and impact at 200 meters ± 5 meters from the weapon into 10 inches, minimum, of coarse sand. The cartridges shall be fired at a rate not to exceed four per minute. At the start of each functioning test six (6) gun warmers shall be fired. Velocities of the cartridges shall not be recorded. The sand shall be raked so that idividual craters do not overlap. In addition, the sand shall be raked after each dud. If a projectile assembly falls outside the impact medium, it shall be declared a "no-test". When the proving ground reports ballistic test results, the number of "no-test" cartridges shall also be recorded. Observation shall be made for functioning on the impact medium and any premature bursts or air bursts. The expended cartridge cases shall be examined for primer leakage and base plug movement. The samples subjected to this test shall also be examined for compliance with base plug security and action time (see 4.5.4).

4.5.10 Heat seal test packing seals. The test shall be conducted in accordance with MIL-P-116.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level A. Preservation and packaging shall be in accordance with Dwg. 9251996 or 9362543 as applicable.

5.2 Packing.



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5.2.1 Level A. The cartridge assembly shall be packed in accordance with Dwg. 9251996 or 9362543 as applicable.

5.3 Marking. Marking shall be in accordance with the applicable drawing and Code of Federal Regulations, Title 49, Parts 100-199.

5.4 Shipping. When shipments of more than one lot are made at one time, such as in a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

5.5 Palletizating. Palletization shall be as described on Dwg. 9362543.

6. NOTES

6.1 Intended use. Not applicable.

6.2. Ordering data. See MIL-A-48078.

6.3. Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARDC, AMSMC-QAF-I(D), Dover, NJ, 07801-5001. Request letter of submittal state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

6.4 Submission of results of contractor conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examination or tests to Commander, ARDC, ATTN: AMSMC-QAF-S(D), Dover, NJ 07801-5001.

6.5 Submission of test data. In addition to the normal distribution of records, when the cartridges are procured by AMCCOM, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commander, ARDC, ATTN: AMSMC-QAF-S(D), Dover, NJ 07801-5001.

6.6 Cost of check test. The Contracting Officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the primer assemblies. The tests shall be conducted at government expense without cost to the contractor who loaded the primer assemblies into the cartridge and shall not constitute a basis for rejection against either contractor except where deterioration has occured as a direct result of carelessness in handling, storage, etc., permitted while the primer assembly lot was under the jurisdiction of either contractor (when applicable).



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6.7 Standard deviation shall be calculated from the following formual:

$$S = \sqrt{\frac{(X - \overline{X})^2}{n-1}}$$

where: $X$ = each individual value

$\overline{X}$ = sample arithmetic mean $\frac{\Sigma X}{n}$

$n$ = sample size

$(X - \overline{X})$ = the sum of the squares of the differenced between the sample mean and each individual value.

6.7.1 For a faster and easier method of arriving at the standard deviation formula, the following alternate method may be used:

$$S = \sqrt{\frac{n \Sigma X^2}{n} - \frac{(\Sigma X)^2}{(n-1)}}$$

6.8 Propellant weight. If any cartridge case, prior to insertion of the base plug and primer, is suspected of containing an underweight or overweight propellant charge, based on visual inspection, it shall be removed from the lot and weighed on a suitable scale. Any cartridge case found with less than 75 percent or more than 125 percent of the assessed propellant weight will be classified as a critical defect and removed from the lot. Any cartridge case found to have a propellant weight out of drawing tolerance but within ± 25 percent of the assessed propellant weight shall be classified as a major defect and removed from the lot.

6.9 Air burst. A projectile that functions in flight further than 20 feet from the weapon muzzle.

6.10 Premature burst. A premature burst is considered to have occured if the projectile functions in the weapon bore or in flight up to 20 feet from the weapon muzzle.

6.11 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or ARRADCOM drawings. Technical data originally prepared by these activities are now under the cognizance of ARDC.

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Preparing activity:
Army-AR

(Project 1310-A373)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-C-48684 | Cartridge, 40MM, M918 TP, LOADING, ASSEBLING AND PACKING |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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